Simulation-Based Comparison of Controlled Interrupted Time Series and Traditional Regression for Policy Evaluation

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Table S1: Calculated n_{sim} formulas for performance measures (The number of simulations was chosen using a pilot run of 100 simulations under the most restrictive setting (n =12,ρ =0.8,effect size(small) =-0.0408), such that the Monte Carlo standard errors for bias, mean squared error, model-based and empirical standard errors were below 0.005, and those for coverage and power were below 0.5 percentage points).

Performance measure	Formula for number simulations required(n_{sim})	Number of simulations(n_{sim})
Bias	$\dfrac{\frac{1}{n_{sim-1}}\sum_{1}^{n_{sim}}(\hat{\theta}_i-\bar{\theta})^2}{\left(\sqrt{\frac{1}{n_{sim}(n_{sim-1})}\sum_{1}^{n_{sim}}(\hat{\theta}_i-\bar{\theta})^2}\right)^2}$	8688
Mean square error (MSE)	$\dfrac{\frac{1}{n_{sim-1}}\sum_{i}^{n_{sim}}\left[(\hat{\theta}_i-\theta)^2-MSE\right]^2}{\left(\sqrt{\frac{1}{n_{sim}(n_{sim-1})}\sum_{1}^{n_{sim}}\left[(\hat{\theta}_i-\theta)^2-MSE\right]^2}\right)^2}$	2827
Empirical standard error(EmpSE)	$1+\dfrac{\widehat{EmpSE}^2}{2\left\{MCSE\left(\widehat{EmpSE}\right)\right\}^2}$	4345
Model-based standard error (ModSE)	$\dfrac{Var\left[\widehat{Var}(\hat{\theta})\right]}{4\widehat{ModSE}^2\left\{MCSE\left(\widehat{ModSE}\right)\right\}^2}$	822
95% Coverage	$\dfrac{Coverage(1-Coverage)}{(MCSE(Coverage))^2}$	1900
80% Power	$\dfrac{Power(1-Power)}{(MCSE(Power))^2}$	6400

Here, MCSE denotes the desired Monte Carlo standard error and ρ denotes lag-1 auto-correlation. Consequently, the largest required number of simulations (8,688) was selected.

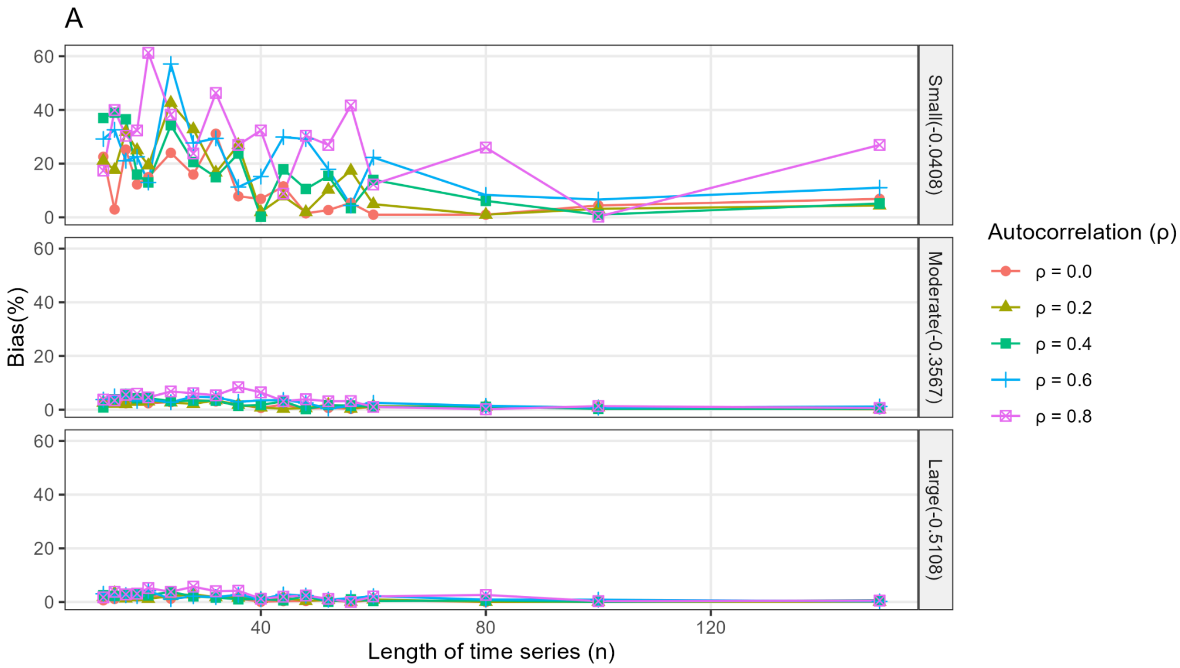
Table S2: Summary of key studies informing parameters of this study

Author	Year	Link
Hammitt et al.	2019	https://doi.org/10.1016/S0140-6736(18)33005-8
Silaba et al.	2019	https://doi.org/10.1016/S2214-109X(18)30491-1
Bottomley, Isham & Scott	2019	https://doi.org/10.1515/em-2018-0010
Lopez Bernal et al	2017	https://pubmed.ncbi.nlm.nih.gov/27283160/
Haeusler et al.	2025	https://doi.org/10.1371/journal.pgph.0004888
Ong'ayo et al.	2019	https://doi.org/10.1016/S2214-109X(19)30188-3
Turner et al.	2021	https://pubmed.ncbi.nlm.nih.gov/34454418/
Talbert et al.	2023	https://pubmed.ncbi.nlm.nih.gov/36631234/
TP Morris, IR White & MJ Crowther	2019	https://doi.org/10.1002/sim.8086

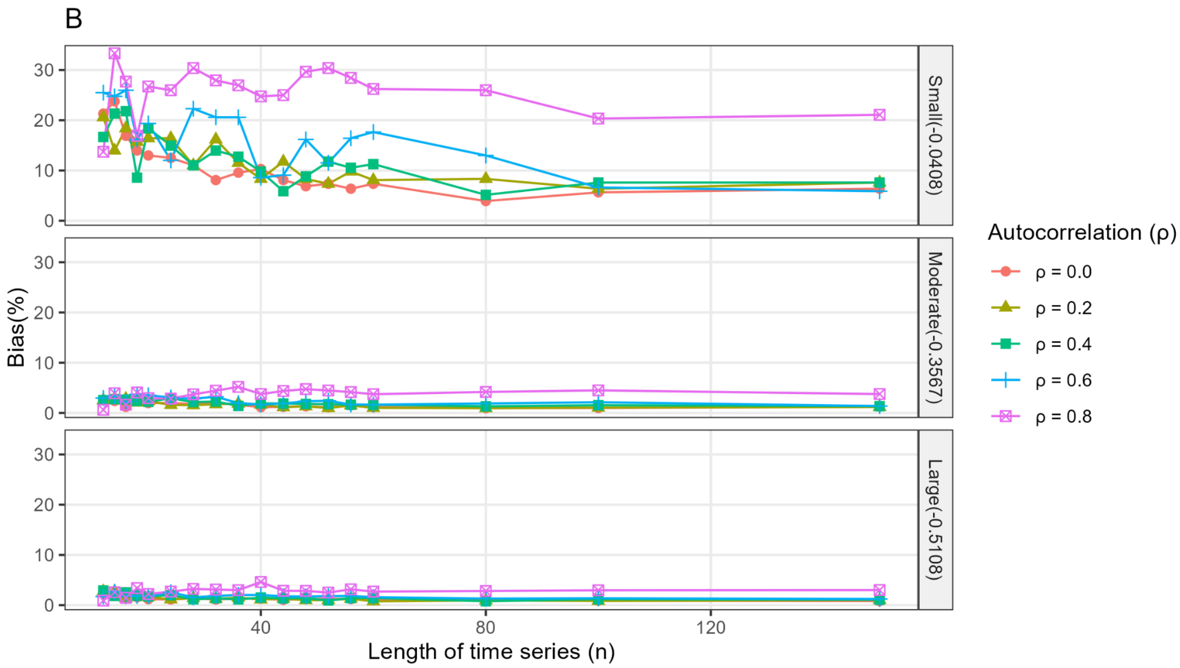
Table S3: Definitions of performance measures.

Performance measure	Definition	Estimate
Bias	$E[\hat{\theta}] - \theta$	$\frac{1}{n_{sim}}\sum_{i=1}^{n_{sim}} \hat{\theta}_i - \theta$
Empirical standard error	$\sqrt{Var(\hat{\theta})}$	$\sqrt{\frac{1}{n_{sim}-1}\sum_{i=1}^{n_{sim}} (\hat{\theta}_i - \bar{\theta})^2}$
Mean square error	$E\left[(\hat{\theta}_i - \theta)^2\right]$	$\frac{1}{n_{sim}}\sum_{i=1}^{n_{sim}} (\hat{\theta}_i - \theta)^2$
Coverage	$Pr(\hat{\theta}_{low} \leq \theta \leq \hat{\theta}_{upp})$	$\frac{1}{n_{sim}}\sum_{i=1}^{n_{sim}} 1(\hat{\theta}_{low,i} \leq \theta \leq \hat{\theta}_{upp,i})$
Power	$Pr(p_i \leq \alpha)$	$\frac{1}{n_{sim}}\sum_{i=1}^{n_{sim}} 1(p_i \leq \alpha)$
model-based standard error	$\sqrt{E\lfloor \hat{Var}(\hat{\theta})\rfloor}$	$\sqrt{\frac{1}{n_{sim}}\sum_{1}^{n_{sim}} \hat{Var}(\hat{\theta}_i)}$

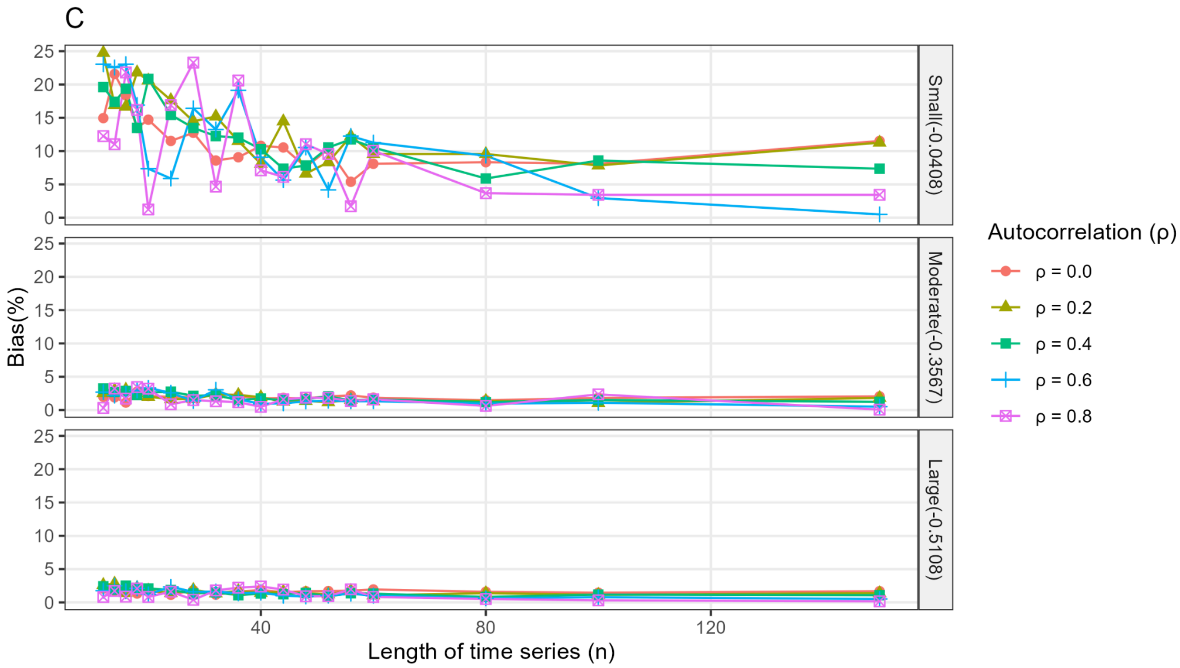
Where θ represents the parameter under investigation, $\hat{\theta}$ being the estimate of that parameter, $\bar{\theta}$ being the mean value of the estimate, n_{sim} being the number of simulations (in this study, 86,88), p_i being the p-value of estimate i and α being the significance level (5%).



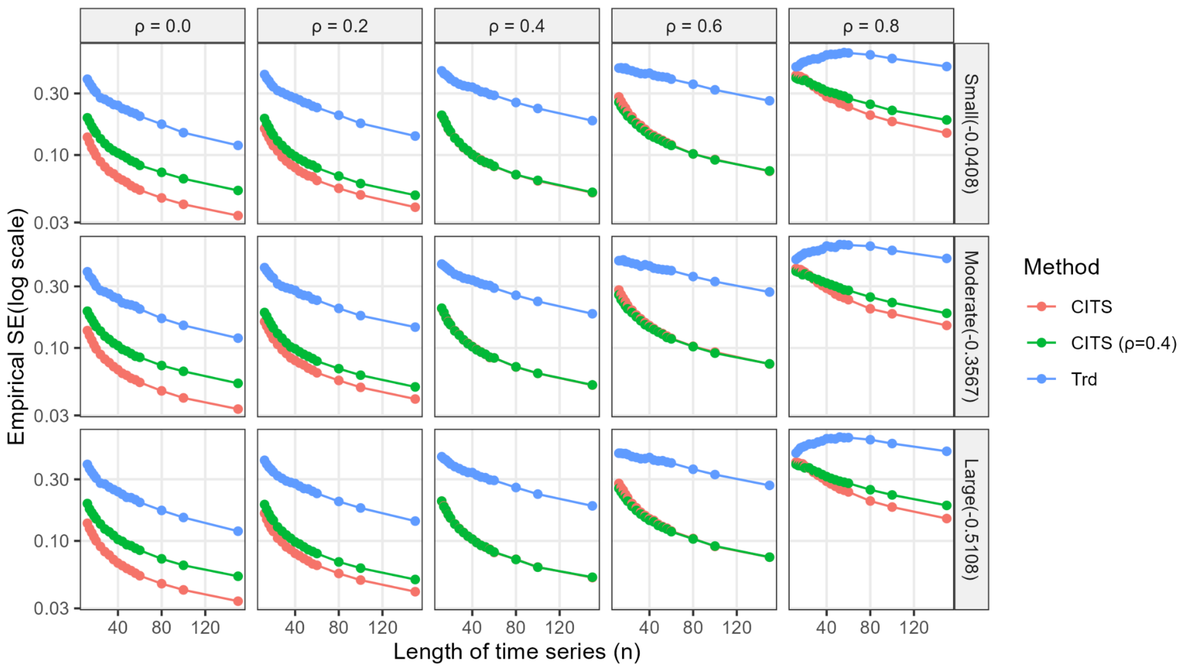
A
Small(-0.0408)
Moderate(-0.3567)
Large(-0.5108)
Bias(%)
Length of time series (n)
Autocorrelation (ρ)
ρ = 0.0
ρ = 0.2
ρ = 0.4
ρ = 0.6
ρ = 0.8



B
Small(-0.0408)
Moderate(-0.3567)
Large(-0.5108)
Bias(%)
Length of time series (n)
Autocorrelation (ρ)
ρ = 0.0
ρ = 0.2
ρ = 0.4
ρ = 0.6
ρ = 0.8



Supplementary Figure 1. Results from the multivariable negative binomial regression (A), controlled interrupted time series model (B), and controlled interrupted time series model with control-series autocorrelation fixed at 0.4 (C).



Supplementary Figure 2. Empirical standard errors from the multivariable negative binomial regression (Trd) and the controlled interrupted time series (CITS) model in which the control series shares the exact lag-1 autocorrelation magnitude (ρ) with the outcome series (Yt), as well as CITS with fixed autocorrelation ($\rho = 0.4$).